

Probabilistically constrained vector summation of motion direction in the mouse superior colliculus

Highlights

- Mouse SC computes vector sum (VS) to integrate motion direction of asymmetric plaids
- The VS is constrained probabilistically by predictions based on single gratings
- Excitatory and inhibitory SC neurons respond similarly to asymmetric plaids
- Probabilistically constrained VS computation also guides optokinetic reflex

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In brief

Li et al. demonstrate that neurons in the mouse superior colliculus compute vector sum when integrating visual motion directions, which is further constrained probabilistically by predictions based on each stimulus component. Such a computation guides optokinetic reflex and differs from what has been shown in visual cortex and visual perception.



Article

Probabilistically constrained vector summation of motion direction in the mouse superior colliculus

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SUMMARY

Visual motion is a crucial cue for the brain to track objects and take appropriate actions, enabling effective interactions with the environment. Here, we study how the superior colliculus (SC) integrates motion information using asymmetric plaids composed of drifting gratings of different directions and speeds. With both *in vivo* electrophysiology and two-photon calcium imaging, we find that mouse SC neurons integrate motion direction by performing vector summation of the component gratings. The computation is constrained probabilistically by the possible physical motions consistent with each grating. Excitatory and inhibitory SC neurons respond similarly to the plaid stimuli. Finally, the probabilistically constrained vector summation also guides optokinetic eye movements. Such a computation is fundamentally different from that in the visual cortex, where motion integration follows the intersection of the constraints. Our studies thus demonstrate a novel neural computation in motion processing and raise intriguing questions regarding its neuronal implementation and functional significance.

INTRODUCTION

How the nervous system processes visual motion information in the environment is critical for the animal's survival, such as in capturing running prey and escaping from looming predators. Using simple stimuli such as bars, dots, and gratings, past studies have discovered that many neurons in the early visual system respond selectively to motion stimuli, especially to motion direction. In natural scenes, processing motion can be challenging because the signals are numerous and complex across the visual field. Proper integration of the motion signals is necessary for effectively interpreting visual stimuli and taking corresponding actions.

Researchers have used the so-called “plaids” to study motion integration, where two gratings of different orientations are superimposed and drifting along their respective axes. Perceptually, human subjects see such stimuli as a rigid plaid pattern with a coherent motion, instead of gratings sliding through each other.¹ Physiological studies showed that plaids are represented in a hierarchical manner in the primate visual cortex, with V1 neurons first decomposing the plaids into grating components and the downstream area middle temporal area (MT) subsequently integrating V1 input to encode pattern motion.² Consistent with perception, the integration of motion direction of MT neurons was found to follow the intersection of constraints (IOC),^{3,4} where all the possible motions given a component grating's motion form a constraint line (Figures 1A–1C, green dashed lines), and the intersection

of the constraints becomes the pattern motion of the plaid (Figures 1A–1C, yellow arrows).

In addition to perception, which guides our understanding and interpretation of the world, motion processing is also important for “action”—the motor output that allows animals to interact with the environment.^{5–9} The dynamic environment that is constantly changing requires an enormous amount of real-time processing that gets updated based on the relationship between the organism and its surroundings. The question is then whether visual processing for perception and action shares the same principle. On the one hand, consistent perception and action would be beneficial for a coordinated biological system. On the other hand, the two functions have different focuses, and a dissociation has been found between them.^{10–12} Specifically, certain illusory visual stimuli were shown to only affect subjects' perception but not their motor action in response to those stimuli.^{13–19}

The superior colliculus (SC) is an evolutionarily conserved midbrain structure critical for action generation.^{20–22} The SC contains visual neurons that are motion direction selective across various mammalian species, and it plays important roles in guiding eye and head movements,^{20,23–25} as well as innate behaviors such as orienting, escaping, and predatory hunting.^{26–30} Recently, direction-selective neurons in the mouse SC were found to respond to the pattern motion rather than component motion of symmetric plaids.³¹ However, this result is consistent with multiple interpretations. For example, other than IOC, vector sum (VS), which is simply the summation of two component



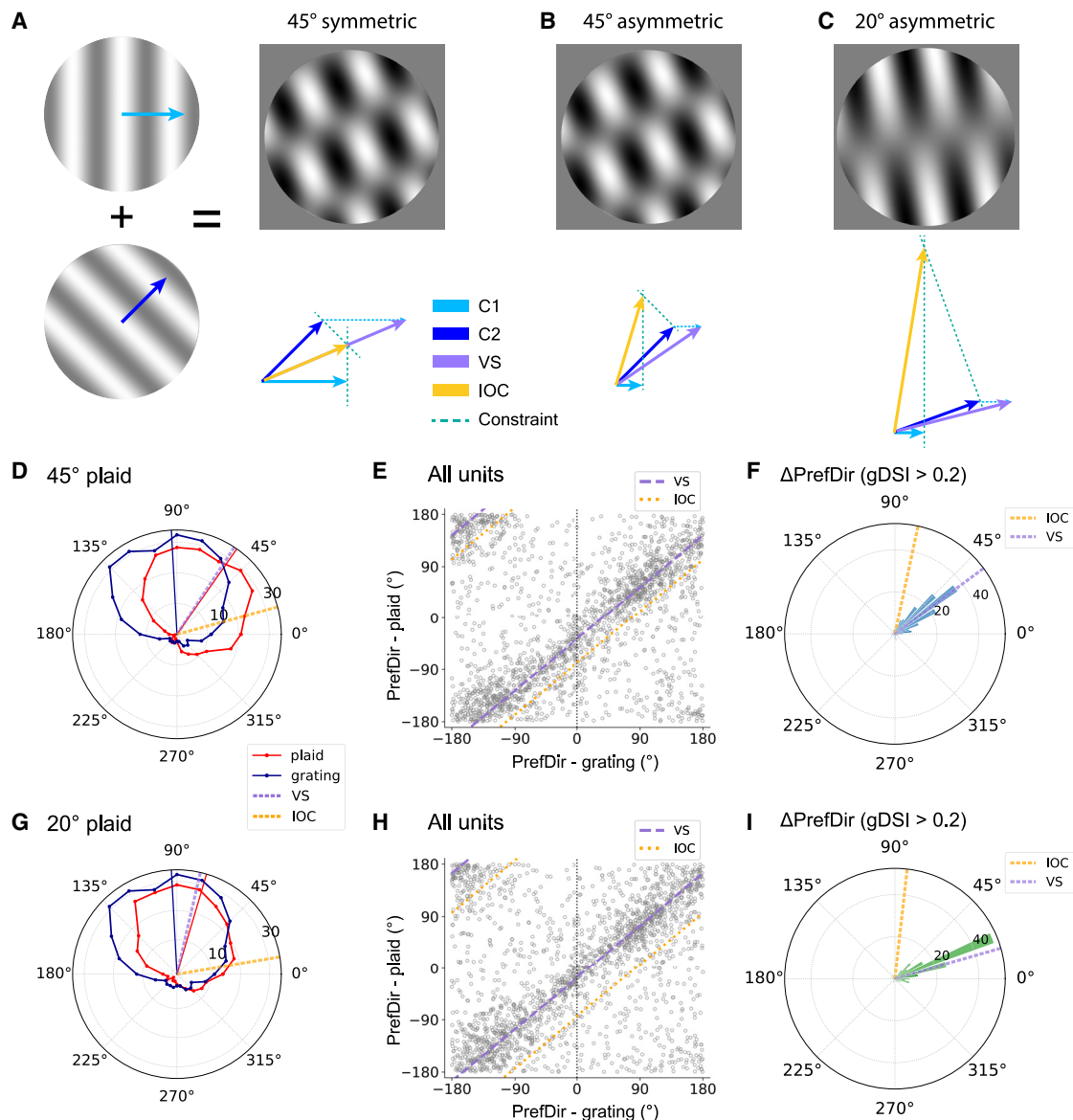


Figure 1. SC performs vector sum for 45° and 20° asymmetric plaids

(A) Schematic of a symmetric plaid with a 45° cross angle. The two component gratings (C1 and C2, light and dark blue, respectively) have the same drifting speed (indicated by the length of the vector). Intersection of constraints (IOC, yellow arrow; intersection of the green dashed lines) and vector sum (VS, purple arrow) have the same direction.

(B) Schematic of an asymmetric plaid with a 45° cross angle. C1 and C2 have a speed ratio of 1:4, and C2 is 45° counter-clockwise to C1. IOC predicts an integrated motion toward the yellow arrow (77.9° counter-clockwise to C1), while VS predicts an integrated motion toward the purple arrow (36.5° counter-clockwise to C1).

(C) Same as (B) but the cross angle is 20°. The IOC direction is 83.6° counter-clockwise to C1. The VS direction is 16.0° counter-clockwise to C1.

(D) An example neuron's tuning to drifting grating (dark blue curve, with preferred direction of 93.1°). Purple and yellow dashed lines are the VS and IOC prediction of the preferred direction to the 45° asymmetric plaid when the tuning curve is plotted in reference to C1's direction. This example neuron's tuning to 45° plaid follows VS (red curve, with preferred direction of 55.0°).

(E) The preferred direction to grating (x axis) versus to 45° plaid (y axis) of all recorded clusters ($n = 1,941$, including both single- and multi-units). Purple and yellow dashed lines are the VS and IOC predictions, respectively.

(F) Difference of the preferred direction between responses to grating and 45° plaid (grating – plaid) for highly selective clusters (gDSI > 0.2 for both grating and plaid responses). Purple and yellow dashed lines are the VS and IOC prediction. The distribution has a mean of 35.6° and a standard deviation of 42.6°.

(G–I) Same as (D)–(F), but for the 20° asymmetric plaid. The example neuron is the same as in (D). The mean of the distribution in (I) is 17.9° with a standard deviation of 40.7°.

See also [Figure S1](#) and [Video S1](#).

motion vectors, predicts the same integrated direction (Figure 1A, purple arrow).

In this study, we aim to investigate the rule of motion integration in the SC by studying its response to asymmetric plaids with unequal component motion strength (e.g., Figure 1B, right). The IOC and VS lead to contrasting predictions when asymmetric plaids are used, thus providing an opportunity to test which computation the SC performs. Surprisingly, we find a clear VS representation in the mouse SC, which is strikingly different from the integration rule in the primate visual cortex. Interestingly, the VS computation in the SC is constrained probabilistically by the possible physical motions consistent with each individual grating. Further, we show that the excitatory and inhibitory SC neurons respond similarly to the plaid stimuli. Finally, by examining optokinetic reflex (OKR), a behavior guided by subcortical circuits, we show that the optokinetic eye movement follows the probabilistically constrained VS computation in response to asymmetric plaids. Therefore, this computation is likely a general principle of motion integration that is shared across visual action pathways.

RESULTS

SC neurons perform VS in response to asymmetric plaids

We recorded visual responses from the superficial layers of the SC in awake mice ($n = 8$) with 64-channel silicon microprobes. Many neurons showed direction-selective responses to single drifting gratings (e.g., Figure S1), as previously reported.^{31–34} To investigate how SC neurons integrate visual motion, we studied their responses to asymmetric plaids. We set the speed of the two component gratings (C1 and C2) to be 6.25°/s and 25°/s, respectively (i.e., ratio of 1:4), and displayed the tuning curves relative to the direction of C1. When C1 and C2 have a cross angle of 45°, their VS is 36.5° away from C1's direction (counter-clockwise in Figure 1B), whereas the IOC direction is 77.9° from C1 (Figure 1B). For the example neuron, which prefers direction of 93.1° to gratings (Figures 1D and S1, dark blue trace), it would prefer a 56.6° (i.e., 93.1°–36.5°) direction in response to the 45° cross angle asymmetric plaid if it follows the VS rule to integrate plaid motion and 15.2° (i.e., 93.1°–77.9°) if it follows the IOC rule. Amazingly, this neuron's preferred direction to the 45° plaid became 55.0°, almost perfectly aligned to the VS (Figures 1D and S1, red trace). The same finding was seen at the population level, including both single units and multi-unit clusters (Figure 1E). We then calculated the difference between the preferred direction to gratings and plaids for all the highly direction-selective units (global direction selectivity index [gDSI] > 0.2). The distribution (mean = 35.6°, SD = 42.6°, $n = 185$, Figure 1F) centered at the VS prediction (i.e., 36.5°), instead of IOC (i.e., 77.9°).

We next used asymmetric plaid with a 20° cross angle, where VS prediction would have a shift of 16.0° and IOC a shift of 83.6° (Figure 1C). The VS and IOC predictions are more separated for the 20° plaid than for 45° plaid, providing an even clearer test of how SC neurons integrate motion. The example neuron was tuned to the VS direction of the 20° plaid rather than IOC (Figure 1G, a shift of preference from 93.1° to 73.5°, i.e., a 19.6° difference), and the same observation was seen at the population

level (Figure 1H). The distribution of preferred direction difference for gratings and plaids for highly selective units was again found to be centered at the VS prediction (Figure 1I, mean = 17.9°, SD = 40.7°, $n = 174$). These results thus demonstrate that SC neurons perform VS computation, rather than IOC, in response to the asymmetric plaids with 20° and 45° cross angles.

SC neurons represent motion direction as a joint probability function in response to asymmetric plaids

The plaids with 20° and 45° cross angles fall into the category of “type II” plaids, where both component directions are on the same side of the IOC direction. We next expanded the cross angle to 135°, which no longer meets the type II definition but still has different VS (shift of 122.9°) and IOC (shift of 81.5°) predictions (Figure 2A). The spatial pattern of a 135° plaid is a mirrored version of 45° plaid (see stimulus video in supplemental information). Surprisingly, in response to the 135° plaid, the same example neuron preferred a direction closer to the IOC prediction than VS (Figure 2B, a shift of preferred direction from 93.1° to 359.8°, i.e., a 93.3° difference). This was also true for the population data (Figure 2C). For highly direction-selective units, the distribution of preferred direction difference between grating and plaids was centered near the IOC prediction (mean = 86.7°, SD = 53.1°, $n = 132$) and showed a wider spread than the distribution for 20° and 45° plaids (Figure 2D).

To understand this puzzling result, we note an important difference between 135° plaid and the other plaids we tested: the VS direction of 135° plaids exceeds the $\pm 90^\circ$ range of C1 direction (Figures 2A and 2E). The $\pm 90^\circ$ range is critical because it is the limit of possible directions of 1-dimensional motion stimuli such as drifting gratings. In other words, a grating drifting along its perpendicular direction could have a “true” motion direction anywhere within the $\pm 90^\circ$ range, but not beyond, because any motion parallel to the grating is invisible when shown in a limited patch (Figure 2E, yellow shade and gray arrows).

Given this property of 1-dimensional motion stimuli, we hypothesized the following computational process. When integrating two gratings of a plaid, the $\pm 90^\circ$ constraints from each component are also combined, thus forming a joint prediction of the shared motion direction in the form of a probability distribution (Figures 2E–2H, “joint prediction,” see STAR Methods for a detailed explanation of the shape of the curve). At the same time, VS of the two components results in a likelihood distribution centered at the VS direction (Figures 2E–2H, “likelihood by VS”). The joint prediction and likelihood distribution are then multiplied to achieve an unnormalized joint probability density distribution of the stimulus direction (Figures 2E–2H, “joint probability density function [pdf]”). We propose that SC neurons' direction tuning in response to plaids is generated following a stochastic process, probabilistically sampling from this joint pdf distribution. When the cross angles are smaller than 90°, the multiplication only has a scaling effect on the direction representation (Figures 2E, 2F, and S2B). For the 135° plaid, however, the peak of the likelihood falls outside of the range of high prediction probability (Figures 2G and S2C). The peak is hence attenuated, while the likelihood within the high prediction region is amplified, resulting in the joint pdf peak away from the VS and closer to the IOC.

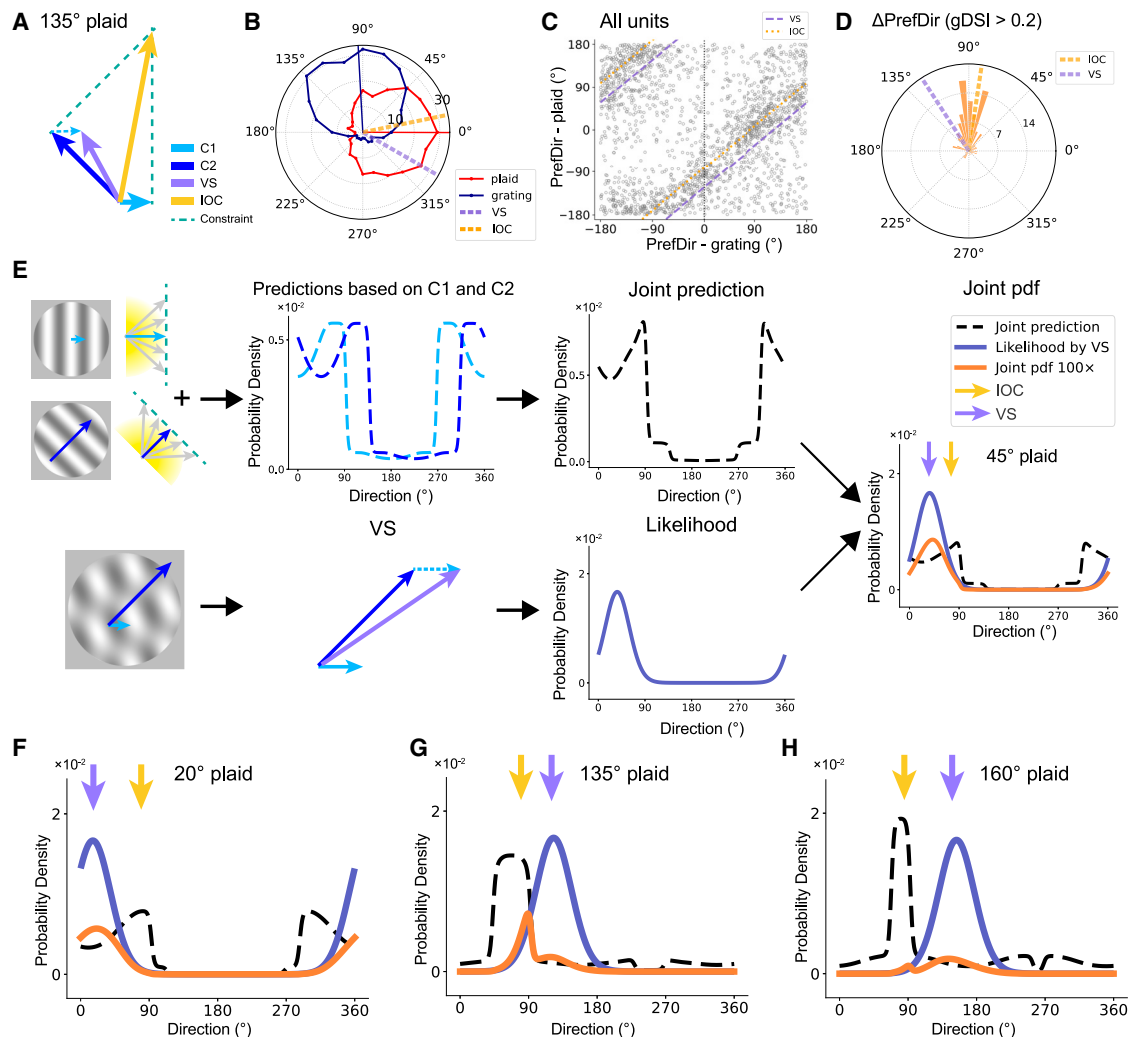


Figure 2. SC response deviates from VS for 135° plaids and can be explained by a probabilistically constrained VS computation

(A) Schematic of the 135° asymmetric plaid. IOC predicts an integrated motion toward the yellow arrow (81.5° counter-clockwise to C1), while VS predicts an integrated motion toward the purple arrow (122.9° counter-clockwise to C1).

(B) The tuning curve of the same example neuron as in Figures 1D and 1G to grating (dark blue curve) and the 135° plaid (red curve). Purple and yellow dashed lines are the VS and IOC predictions, respectively. The neuron's preferred direction to the plaid (359.8°) is closer to IOC than to VS.

(C) The preferred direction to grating versus to 135° plaid of all recorded clusters ($n = 1,941$). Purple and yellow dashed lines are the VS and IOC predictions, respectively.

(D) Difference of the preferred direction between responses to grating and 135° plaid for highly selective clusters ($\text{gDSI} > 0.2$). The distribution has a mean of 86.7° and standard deviation of 53.1°.

(E) Top, given the motion of a single grating, the possible true motion lies within its $\pm 90^\circ$ range (yellow shades). Gray arrows indicate some example motion vectors that are compatible with the grating. The possible true motion of each grating is modeled as a probability density function (pdf) as illustrated by the light and dark blue dashed lines, respectively. The normalized product of the two functions is the joint prediction function (black dashed line). Bottom, VS is modeled with a likelihood distribution (von Mises function) centered at the VS direction (purple line). Multiplication of the joint prediction and the likelihood function returns the final joint probability function (orange). Purple and yellow arrows are the VS and IOC predicted motion direction for the 45° plaid.

(F) Same as the joint pdf as in (E) but for 20° plaid.

(G) Same as (F) but for 135° plaid. In contrast to the 45° and 20° plaids, the peak of the likelihood distribution (purple curve) for the 135° plaid falls near the prediction decay region (cliff of the black dashed line). The VS peak is attenuated after multiplication with the prediction distribution, resulting in a joint pdf peak away from VS but closer to IOC (orange curve, peak closer to the yellow arrow than purple).

(H) Same as (F) but the prediction of SC response for 160° plaid. The likelihood (purple curve) of 160° plaid falls completely out of the prediction transition region. The resulting joint pdf (orange) maintains its peak at the VS direction (purple arrow). Due to the multiplication with low prediction values, the likelihood turns into a joint pdf distribution with a much lower amplitude.

See also Figure S2 and Video S1.

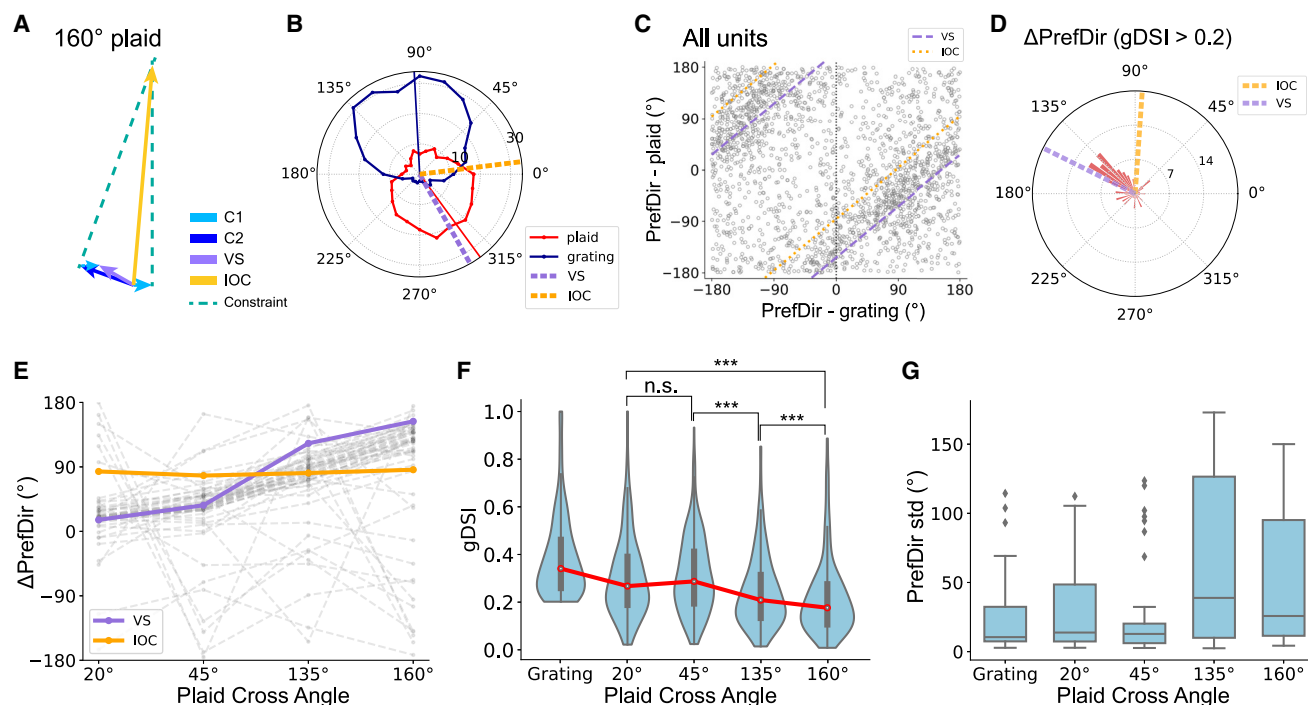


Figure 3. SC performs probabilistically constrained VS to represent the direction of asymmetric plaids

(A) Schematic of the 160° asymmetric plaid. IOC predicts an integrated motion toward the yellow arrow (86.0° counter-clockwise to C1), while VS predicts toward the purple arrow (153.6° counter-clockwise to C1).

(B) The tuning curve of the same example neuron as in Figures 1D, 1G, and 2B to grating (dark blue curve) and the 160° plaid (red curve). The neuron's preferred direction to the 160° plaid (306.1°) is closer to VS (purple dashed line) than to IOC (yellow dashed lines).

(C) The preferred direction to the grating versus to 160° plaid of all recorded clusters ($n = 1,941$). Purple and yellow dashed lines are the VS and IOC predictions, respectively.

(D) Difference of the preferred direction between responses to the grating and 160° plaid for highly selective clusters ($\text{gDSI} > 0.2$). Purple and yellow dashed lines are the VS and IOC predictions, respectively. The distribution has a mean of 139.7° and standard deviation of 71.3°.

(E) Difference of the preferred direction between responses to the grating and plaids for clusters that are highly selective ($\text{gDSI} > 0.2$) for all 4 plaid conditions ($n = 61$). Gray dashed lines connect responses of individual units. Purple and yellow lines are the VS and IOC predictions, respectively.

(F) The gDSI of clusters that are highly direction selective ($\text{gDSI} > 0.2$) to grating ($n = 261$) under all 5 stimulus conditions. Friedman test, $p < 0.001$. Post hoc Wilcoxon tests are performed between adjacent plaid conditions, $p = 0.76$ for 20° and 45° plaids, $p < 0.001$ for 45° and 135° plaids, and for 135° and 160° plaids. The connected points represent the median. The thick vertical bars within each column represent the range from the first to the third quartiles. The whiskers represent the data range excluding the outliers.

(G) The standard deviation of the preferred direction estimation across 10 stimulus repeats for individual neurons. In response to the 20° (SD median = 13.80°) and 45° (SD median = 12.81°) plaids, the preferred direction estimates are comparably reliable as to gratings (SD median = 10.49°). For plaids with larger cross angles, the variability increases (SD median = 38.86° for 135° plaid; SD median = 25.77° for 160° plaid). The horizontal lines in each box represent the median. The boxes represent the range from the first to the third quartiles. The whiskers represent the data range excluding the outliers. The diamond points are outliers. See also Video S1.

Such a multiplicative operation of pdfs makes a specific prediction: if the cross angle is expanded further, to 160°, for example, where the likelihood by VS falls in the low but flat prediction region, the SC responses would again become closer to the VS prediction, but weaker in strength due to a more dispersed joint pdf (Figures 2H and S2D). We tested this prediction by investigating SC responses to 160° plaids, where VS predicts a shift of 153.6° and IOC predicts a shift of 86.0°. Indeed, in response to these plaids, the direction representation was closer to the VS than IOC (Figures 3A–3D, example neuron tuning shifted 147.0°; Figure 3D: mean = 139.7°, SD = 71.3°, $n = 110$). We next compared responses of the same units that were highly selective across all 4 plaid conditions. Their response shifted from VS for 20° and 45° plaids to IOC-like for 135° plaids, then back to VS for 160° plaids (Figure 3E). Remarkably, as predicted

by the joint pdf, fewer units showed highly selective response for plaids with larger cross angles ($n = 174, 185, 132$, and 110, for 20°, 45°, 135°, and 160° plaids, $\text{gDSI} > 0.2$). Consistently, the direction selectivity across population also became weaker with increasing cross angles (Figure 3F, $n = 261$, $\text{gDSI} > 0.2$ to gratings). Moreover, the preferred direction estimated from each repeat of stimuli showed a higher variability for plaids with larger cross angles (Figure 3G), also consistent with the probabilistically constrained VS model.

Excitatory and inhibitory SC neurons respond similarly to asymmetric plaids

We next performed two-photon calcium imaging to target direction-selective neurons that are enriched in the superficial SC,^{34,35} which also allowed us to study specific neuron types

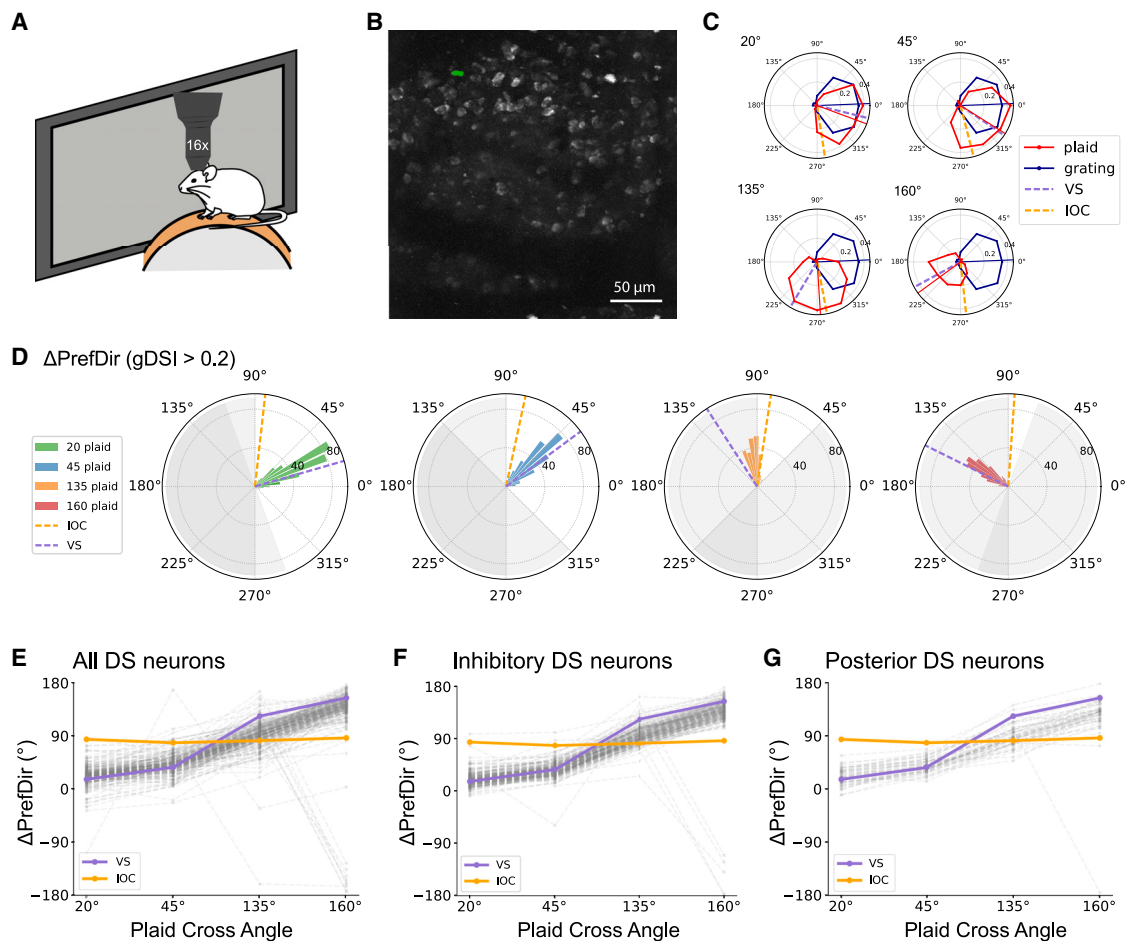


Figure 4. Excitatory and inhibitory SC neurons respond similarly to asymmetric plaids

(A) Schematic of two-photon imaging experiment. Head-fixed mice were allowed to freely run on a wheel while passively viewing the visual stimulus. (B) An example field of view imaged from the anterior SC. Both excitatory and inhibitory neurons were included. Green marks an example neuron. (C) The example neuron's tuning curve to grating and plaids of 4 different cross angles. This neuron's preferred direction was 2.2° for grating and 339.7°, 327.9°, 273.6°, and 216.2° for 20°, 45°, 135°, and 160° plaids, respectively (i.e., a shift of 22.5°, 34.3°, 88.6°, and 146.0°). (D) Distributions of the difference between preferred directions of responses to the grating and plaids for all direction-selective (DS, $gDSI > 0.2$) neurons imaged from the anterior SC. (E) The preferred direction difference between responses to the grating and plaids for all direction-selective neurons imaged from the anterior SC ($gDSI > 0.2$). Gray dashed lines connect individual neurons. Purple and yellow solid lines are VS and IOC predictions, respectively. (F) Same as (E), but for inhibitory neurons only. (G) Same as (E), but for DS neurons imaged from the posterior SC. See also Figure S3.

(Figures 4A and 4B). First, we injected adeno-associated virus (AAV) to express GCaMP6s in both excitatory and inhibitory neurons in similar locations as in the electrophysiological recording experiments ($n = 3$ mice). Overlaying cortical tissues were removed to expose the SC surface for imaging.³¹ In response to the same 4 types of plaid stimuli, the imaged neurons shifted their direction preference exactly like what was observed with electrophysiology—VS for 20°, 45°, and 160° plaids and closer to IOC for 135° plaid (Figures 4C and 4D). The same results held true for neurons that were highly selective across all 4 plaid conditions ($n = 191$, Figure 4E).

Next, we injected AAV to express Cre-dependent GCaMP6s in Vgat-Cre mice, resulting in GCaMP6s expression only in inhibitory neurons ($n = 3$ mice). A VS representation was again

observed for 20°, 45°, and 160° plaids, whereas IOC-like representation was seen for 135° plaids, suggesting that inhibitory and excitatory neurons respond similarly to these plaids (Figure 4F).

Finally, to determine if this newly discovered response property displayed any regional specificity, we imaged the posterior and medial corner of the SC that has receptive fields more lateral and dorsal in visual field ($n = 2$ mice). Notably, no tissue was lesioned in this experiment, leaving cortical input intact.³⁶ The same response pattern to plaids was also observed for direction-selective neurons in this region of the SC (Figure 4G). Additionally, in all three sets of imaging experiments, the direction selectivity showed a decreasing trend with larger cross angles of plaids (Figure S3), which further confirmed the findings obtained with electrophysiology (Figure 3F).

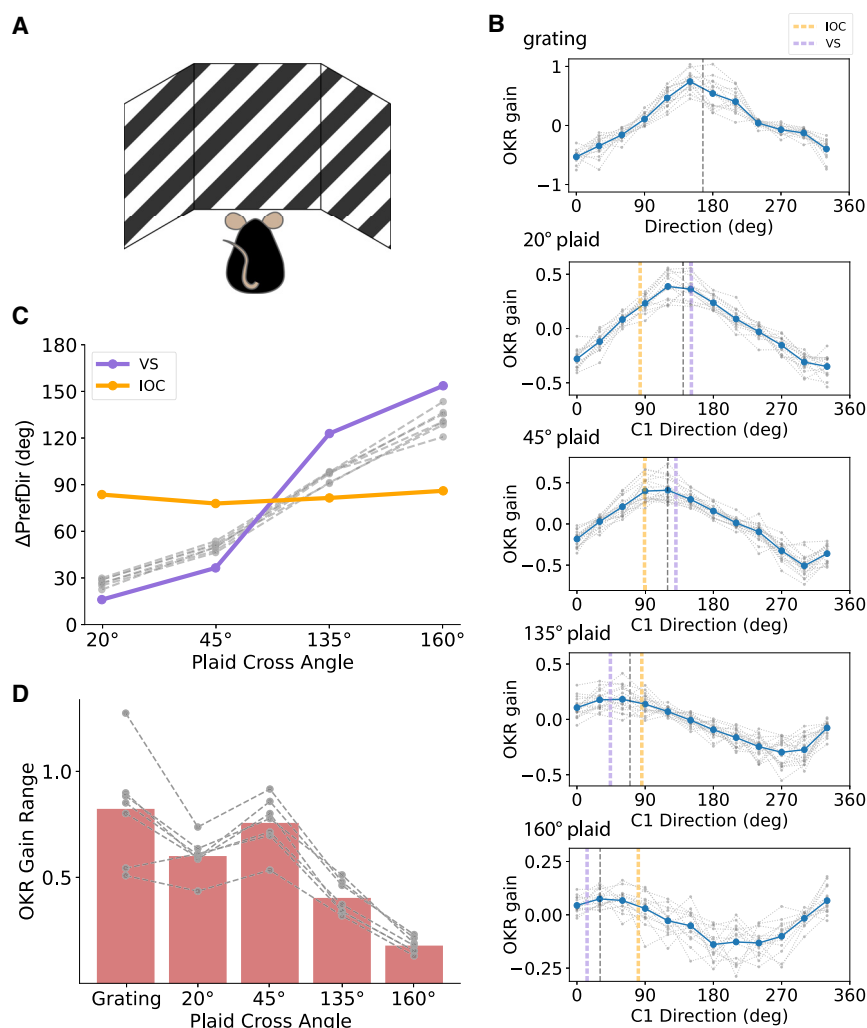


Figure 5. Probabilistically constrained VS computation guides optokinetic eye movement

(A) A schematic of the full-field stimulus delivered by a 3-screen setup.

(B) OKR to grating and all 4 plaids of an example mouse. The x axis is the grating direction (top) or component 1 direction in plaids. Gray dashed lines are individual repeats (15 total for each condition) of the stimulus. Yellow and purple dashed lines in the plaid panels are IOC and VS predictions based on the OKR tuning to grating.

(C) The difference between preferred directions to grating and plaids for all animals ($n = 7$). Gray dashed lines connect data from individual animals. Purple and yellow solid lines are VS and IOC predictions, respectively.

(D) The OKR gain range, calculated as the difference of maximum and minimum OKR gain, for grating and all 4 plaid conditions.

(approximately 8°/s in VS direction, see STAR Methods) were shown on a 3-screen setup surrounding the animal ($n = 7$, Figure 5A). We recorded the right eye through the reflection by a dichroic mirror and measured the speed of its movement along the horizontal axis. As expected, horizontally drifting gratings elicited the largest horizontal OKR response (Figure 5B, top), with 0° grating (drifting leftward) and 180° grating (rightward) evoking eye movements in the corresponding directions (quantified as negative and positive OKR gain, respectively).

We then investigated OKR in response to asymmetric plaids of the 4 cross angles

Together, the robustness of our findings, seen with both physiology and imaging and across SC regions and cell types, indicates that the probabilistically constrained VS is a fundamental computation performed by the mouse SC when integrating motion information.

Probabilistically constrained VS computation guides optokinetic eye movement

Our results demonstrate that the motion integration principle in the SC is strikingly different from that in the cortical pathway. Given the SC's role in reflexive behavior and controlling eye movements, we tested whether there is any behavioral relevance of the subcortical VS representation of motion direction. We turned to OKR. OKR is an important subcortical reflexive behavior where the eye movements follow the stimulus involuntarily and presumably help stabilize images on the retina. Although the SC is not directly involved in generating the slow and smooth movement in OKR,³⁷ we aimed to test whether the VS computation is utilized in this subcortical behavior, without implicating a driving role of the SC.

Full-field drifting gratings (9.375°/s with a spatial frequency of 0.16 cyc/deg and temporal frequency of 1.5 Hz) and plaids

tested in physiological and imaging experiments. If OKR follows the VS direction of the plaid, the stimulus that evokes the largest rightward eye movements (i.e., the peak of the tuning curve) would shift from near-180° to the VS prediction line (Figure 5B, purple dashed lines); whereas if it follows the IOC direction of the plaid, the peak would shift to the IOC prediction line (Figure 4B, yellow dashed lines). Remarkably, the OKR was closer to the VS prediction for 20°, 45°, and 160° plaids, but to the IOC prediction for 135° plaid (Figures 5B and 5C), just like the SC responses. In addition, the strength of OKR responses (OKR gain) became weaker for the 135° and 160° plaids (Figure 5D), consistent with the finding at the neuronal level (Figures 3F, S3B, S3D, and S3F). These results thus suggest that the probabilistically constrained VS is an essential motion processing computation used in the mouse brain to guide behavior.

DISCUSSION

In this study, we found that neurons in the mouse SC integrate plaid motion by computing the VS of the component gratings. This representation is probabilistically constrained by predictions based on individual components of the plaid. These

findings reveal a novel computation of motion transformation that is fundamentally different from what has been found in the primate cortical pathway, where a computation that generates IOC-like results is the dominant principle.^{3,4}

Plaids have been used to study motion integration along visual pathways. Perceptually, plaids are coherently moving “blobs.” On the neuronal level, studies have found V1 represents moving plaids as a linear summation of two gratings (i.e., responding to individual components), while the downstream MT area represents plaids as single patterns in motion.² In other words, the MT signals resemble more of the perceptual end product. Moreover, using unikinetic plaids (a type of asymmetric plaids with one of the components being static), researchers found that the motion represented by MT neurons followed the IOC direction,^{3,4} which is also consistent with perception.^{1,2,38–40} Interestingly, behavioral responses along the VS direction have also been reported in response to certain asymmetric plaids in humans and nonhuman primates, which were sometimes characterized as “misperception.”^{39,41–45} For example, in a direction-matching task where human subjects reported the plaid direction with a pointer, the perceived direction was found to depend on the duration of the asymmetric plaids.⁴² A response along the VS direction was more likely to happen if the plaids were presented for a short duration (60 ms). With longer durations, the perception (measured by direction matching or eye tracking) was found to change from the VS back to IOC.^{39,42,46} In addition, it was found that pigeons followed the VS direction when performing a direction discrimination task in response to asymmetric plaids, measured by pecking between two choices.⁴⁷ Here we show, for the first time, a VS computation at the neuronal level in response to asymmetric plaids. The VS responses of SC direction-selective neurons may be the neural basis underlying the VS-guided behavioral responses in the direction discrimination tasks. The potential role of the SC in such perceptual responses is supported by its short response latency^{48,49} and involvement in many cognitive functions.^{21,22,50–52}

Importantly, our data further demonstrate that not only does the SC represent the VS direction of plaid patterns, but also the optokinetic eye movement follows the VS direction. Together, these suggest that VS may be a behaviorally relevant computational principle shared across multiple subcortical areas, and it may be used for tasks that are fundamental to these areas. The SC is known for its role in driving actions like head and eye movements across species.^{20,23–25} When presented with two possible visual targets, eye movements typically follow a vector average of the stimuli before the winner-take-all selection of a single target.^{53–58} The SC was shown to contribute to both phases of eye movements.^{59–63} This vector averaging, which is a form of weighted VS, shares a very similar computation as VS, and it is beneficial for bringing the subject’s view to the most information-dense area in the visual space. In addition, recent studies in monkeys and mice indicate that the SC is also involved in spatial attention.^{64,65} Together with its role in driving eye and head movements, the SC must compute signals to guide fast-orienting behaviors to maximize information gain. This process is closely related to action generation rather than visual perception; thus it requires a high speed but can tolerate a low precision, which may benefit from a simple operation like VS.

Interestingly, previous studies in humans showed that unikinetic plaids elicited ocular following (the open-loop phase of OKR in primates), with the direction shifting from component direction (the same as VS direction in this case) in ultra-short latency to the 2D pattern direction (consistent with IOC direction) later in time.⁶⁶ Another study compared human closed-loop phase OKR with that of cats in response to plaids and showed that both species follow a direction between the component and pattern directions, but human OKR is biased toward the pattern direction while cat OKR is biased toward the component direction.⁶⁷ In this study, we found that mouse OKR is also not perfectly VS but deviates slightly toward the IOC direction. These results thus suggest that OKR is likely mediated by the competing signals of component (likely a VS) and pattern representations. The relative contribution of the two signals varies on a spectrum across species. OKR is largely controlled by subcortical circuits for afoveate animals, while the control shifts to the cortex in foveate animals.^{67–69} Thus, the position of each species on such component-pattern spectrum might be determined by the subcortical-cortical balance involved in the behavior.

In this study, we also show that the VS computation in the SC is under an additional constraint. VS by itself is a straightforward linear operation that does not depend on any assumptions or constraints. Yet, the constraint as illustrated in our model (joint prediction, [Figures 2](#) and [S2](#)) relies on two assumptions that are closely related to IOC and the perception of object motion. First, to generate the predictions based on individual gratings ([Figure 2E](#), yellow shades and “predictions”), an underlying assumption must be met, which is that a 1D grating pattern is a part of a larger moving pattern viewed through an aperture. Without additional information, the computation does not reject other possible motion directions that are compatible with this 1D moving pattern. On top of that, the combination of the two predictions from component gratings (i.e., the joint prediction) indicates a second assumption, which is that the two gratings come from the same rigid, coherently moving pattern. Only under this assumption does it matter to search for a shared motion that is compatible with both constraints. These two assumptions are the same ones behind IOC, which are also consistent with the perception of plaids. These assumptions suggest an existing knowledge (“prior” in the Bayesian framework) about motion information in the brain. More specifically, the underlying neural network must have mechanisms to deal with the information ambiguity caused by spatially limited receptive fields. Further, the network seems to prioritize processing visual information superimposed spatially as coming from the same entity with locally rigid motion, rather than independent transparent motion.

What neuronal mechanism can give rise to this newly discovered computation in the SC? We recently showed that mouse SC neurons display pattern motion-selective responses to symmetric plaids, and such responses can arise from a nonlinear transformation of converging retinal inputs, which are more diverse in their pattern versus component motion selectivity.³¹ How retinal ganglion cells respond to asymmetric plaids as we used here needs to be studied. A VS computation, and maybe even the constraint, may already exist in the retina, which is then further transformed in the retinocollicular convergence. Additionally, or alternatively, local circuits in the SC may perform VS computation or direction-specific constraint through

weighted connections among cells with different preferred directions. For example, to represent the probable and improbable motions given a single grating, direction-selective neurons may form stronger excitatory connections with similarly tuned neurons and weak excitatory connections (and/or stronger inhibitory connections) with oppositely tuned neurons. It should be noted that the constraint representation should only depend on the stimulus direction, not the motion speed, while VS computation needs to take the speed into account. Consequently, to implement this computation may require the multiplication of two separate streams of stimulus representation. Future studies are needed to understand the underlying circuitry for plaid motion integration in the SC.

Finally, we did not test how changes of other stimulus parameters, such as spatial frequency and contrast, could influence the neural and behavioral response and whether the influence could be explained by the model. In addition, we focused on predicting the direction-selective responses of SC neurons, without considering their tuning to speeds. Thus, our model cannot distinguish VS and vector average, since they predict the exact same direction. Further studies to address this issue will benefit especially from studying neuron populations that have well-characterized speed-tuning function. It is important to note that vector average and VS are not mutually exclusive, as vector average is a special form of weighted VS. A generalized investigation to measure the weights of VS will be ideal in the future.

RESOURCE AVAILABILITY

Lead contact

Further information and requests for resources should be directed to and be fulfilled by the lead contact, Jianhua Cang (cang@virginia.edu).

Materials availability

This study did not generate any unique reagents.

Data and code availability

- Data for electrophysiology, imaging, and behavior experiments have been deposited at Zenodo (10.5281/zenodo.14363067).
- All original code has been deposited at GitHub (<https://github.com/Cang-Lab/Li-2024>).
- Any additional information required to reanalyze the data reported in this paper is available from the [lead contact](#) upon request.

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AUTHOR CONTRIBUTIONS

C.L. and J.C. designed the experiments. C.L. performed electrophysiology and OKR and data analysis with assistance from H.C. and S.T. V.J.D. performed two-photon imaging and data analysis with assistance from H.C. and C.L. C.L. and J.C. wrote the manuscript with input from all authors.

DECLARATION OF INTERESTS

The authors declare no competing interests.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

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SUPPLEMENTAL INFORMATION

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Bacterial and virus strains		
AAV1-hsyn-GCaMP6s	Addgene ⁷⁰	#100843-AAV1
AAV.PHP.eB-hsyn-RiboL1-jGCaMP8s	gift from Kristian Lensjø Lab ⁷¹	N/A
AAV9-hsyn-Flex-GCaMP6s	Addgene ⁷⁰	#100845-AAV9
Experimental models: Organisms/strains		
Mouse: C57BL/6	The Jackson Laboratory	RRID: IMSR_JAX: 000664
Mouse: Vgat-Cre	The Jackson Laboratory	RRID: IMSR_JAX:028862
Deposited data		
Physiology, imaging and behavior data	This paper	Zenodo Data: https://doi.org/10.5281/zenodo.14363068
Software and algorithms		
Psychophysics Toolbox	Psychtoolbox	http://psychtoolbox.org/docs/Psychtoolbox; RRID: SCR_002881
Matlab R2023a	MathWorks	RRID: SCR_001622
MountainSort	Chung et al. ⁷²	RRID: SCR_017446
Custom Written Python Code	This paper	Github: https://github.com/Cang-Lab/Li-2024
Other		
UCLA Silicon Microprobes	Yang et al. ⁷³	Masmanidis Lab: https://masmanidislab.neurobio.ucla.edu/technology.html
Data acquisition	Intan Technologies	RHD Evaluation System

EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS

We used 8 adult wild type C57BL/6 mice for electrophysiology recording of SC visual responses (female=4, male=4, 4-7 months old. The Jackson Laboratory, RRIC, IMSR_JAX:000664). For two-photon calcium imaging, 8 adult Vgat-Cre mice (all male, 5-7 months old. The Jackson Laboratory, stock #028862, RRID, IMSR_JAX:028862) were used. For OKR behavior test, 7 adult C57BL/6 mice (female=5, male=2, 3-4 months old for 3 females and all males, 9 months old for 2 females. The Jackson Laboratory, RRIC, IMSR_JAX:000664) were used. All experimental procedures in this and following sections were approved by the University of Virginia Institutional Animal Care and Use Committee and in conformance with the National Institutes of Health Guide for the Care and Use of Laboratory Animals and the Public Health Service Policy.

METHOD DETAILS

Electrophysiology

Animals were first prepared for head plate implantation and craniotomy. A 2mm diameter cranial window was made 1mm lateral (right side) and 0.5mm anterior to the lambda point. We used 2.5% agarose and Kwik-Cast sealant to cover and seal the craniotomy. A titanium head plate was cemented with C&B Metabond onto the skull. Isoflurane was used for anesthesia during the surgery (2-3%), and carprofen (5mg/kg) was given as analgesic. Eye ointment was applied to the eyes for protection during surgery.

1-4 days after the surgery, electrical signals from the SC of head-fixed awake mice were recorded using 64 channel silicon microprobes⁷³ via a multichannel amplifier from Intan Technologies at 20 kHz. The mice were allowed to freely run on a cylindrical wheel. The receptive field location of each recording session was estimated using flashing dark squares (5° × 5°) across the screen. The grating and plaid stimuli were then centered on the approximate receptive field location. Spikes were bandpass filtered with 300–6000 Hz range. Spike waveforms were sorted offline into single- and multi-units using MountainSort.⁷²

Visual stimuli were generated by customized script in MATLAB based on Psychtoolbox-3 and shown on an LCD monitor (59.7 × 33.6 cm, 60 Hz refresh rate, gamma corrected). Gratings and plaids (100% contrast) were shown within a circular patch (diameter = 40°) against a grey background. The screen was placed 25 cm away from the eye contralateral to the recording site. The grating had 0.08 cpd spatial frequency (SF) and 2 Hz temporal frequency (TF), thus resulting in a speed of 25 deg/s. We used 24 directions evenly

spaced from 0° to 345° with a 15° increment. The 0° direction represents leftward motion of vertically oriented grating. For plaid, both grating components had a SF of 0.08 cpd. Component 1 (C1) was always the slower component with a TF of 0.5 Hz (speed of 6.25 deg/sec), while component 2 (C2)'s TF was 2 Hz (25 deg/s). We varied C1's direction across 24 directions evenly spaced from 0° to 345°. C2 direction was always C1 direction plus cross angle. Four different cross angles were tested: 20°, 45°, 135°, and 160°. The stimulus was presented for 2s with a 0.5s inter-stimulus grey screen. Each condition was repeated 10 times in a pseudorandom order. Movies of the example visual stimuli are provided in [Video S1](#).

Two-photon Calcium Imaging

The surgical and imaging procedures were as described in the previous studies.^{31,36} To image the anterior portion of the SC, a 3.2 mm craniotomy was made on the left hemisphere centered 1.5–2 mm lateral and anterior to the lambda point. The cortex above the SC was aspirated to expose the SC. To image the posterior SC, a 2.5 mm craniotomy was centered over the lambda point. The dura over the colliculus was cut with a sharp needle (30 gauge) to expose the SC surface without lesioning the cortex or the overlying blood vessels. ~50 nl of AAV solution was pressure injected via a glass pipette fitted to an injector (Drummond Scientific Nanoject II) at 500 μ m and 250 μ m below the pia surface of SC.

AAV1-hsyn-GCaMP6s (Addgene #100843⁷⁰) was injected for imaging all neurons (i.e. excitatory and inhibitory; n=2 mice for anterior SC and n=2 for posterior SC). In another mouse, AAV.PHP.eB-hsyn-RiboL1-jGCaMP8s (gift from Kristian Lensjø Lab⁷¹) was used for imaging the anterior SC. No significant difference was seen in direction selectivity between using the two viruses (Mann-Whitney U test, $Z = -1.19$, $p = 0.23$ for grating; $Z = -1.16$, $p = 0.24$ for 20° plaids; $Z = 0.86$, $p = 0.39$ for 45° plaids; $Z = 0.41$, $p = 0.68$ for 135° plaids; $Z = -0.89$, $p = 0.37$ for 160° plaids). We thus combined them in further analysis (i.e. n=3 for recording all neurons from the anterior SC). For imaging specifically the inhibitory neurons in the anterior SC, AAV9-hsyn-Flex-GCaMP6s (Addgene #100845⁷⁰) was injected (n=3).

Mice were imaged 15–21 days post-surgery with a two-photon scanning microscope (Ultima Investigator or Ultima 2Pplus, Bruker Nano Surface Division) with a Ti:sapphire laser (Chameleon Discovery with TPC, Coherent) tuned to a wavelength of 920 nm through a 16 $\times$, 0.8 NA Nikon objective. Imaging data were acquired at 30Hz rate with the PrairieView software 5.4 with a resonant scanner at 2.25 $\times$ optical zoom, resulting in a 412.2 $\times$ 412.2 μ m field of view (at 512 $\times$ 512 pixel resolution). Images averaged over 4 frames were then used for later analysis.

Regions of interest (ROIs) were manually drawn on the motion-corrected average image of a population of neurons over the collected time-series. The raw Ca^{2+} signal of each ROI was calculated by averaging all pixels within that ROI in each frame. Neuronal response was calculated as $\Delta F/F_0 = (F - F_0)/F_0$, where F_0 was the mean of the baseline signal over 0.8s period before stimulus onset, and F was the average fluorescence signal over the 2s stimulus presentation. The mean value of $\Delta F/F_0$ for each stimulus condition was then used for subsequent data analysis for all the neurons.

The visual stimulus used for imaging was similar as in electrophysiology, except for a longer inter-stimulus interval of 3s due to the slow Ca^{2+} dynamics. In addition, 12 directions (0–330° with 30° spacing) were used for each type of stimulus and each condition was repeated 5 times in a pseudorandom order.

Optokinetic reflex (OKR)

Mice were first implanted with titanium head plates cemented with C&B Metabond onto the skull. Isoflurane was used for anesthesia, and carprofen was given as analgesic. Each animal was head-fixed during the OKR test and allowed to run freely on a wheel.

Three computer screens (52.1 $\times$ 29.2 cm for each) were aligned vertically in portrait mode surrounding the animal. The angle between two neighboring screens was ~120°. The distance between the eyes of the animal and the center of the screens was 25 cm. We recorded the mice in the dark with an infrared light source. A *dichroic* mirror was placed in front of the right eye, and a video camera captured the eye from the back of the animal through the reflection in the mirror, thus minimizing visual occlusion. The camera was model FL3-U3-13Y3M-C (FLIR), combined with an Edmund VZM 200i zoom imaging lens. Videos were recorded at 60Hz rate. We used Facemap⁷⁴ to track the pupil location with the centroids coordinates. Only horizontal coordinates were used in our analysis to avoid the issue of direction-dependent gain difference in OKR behavior. Saccade-like eye movements were detected by their rapid speed and replaced by linear interpolation so that only slow eye movements were used to quantify OKR.⁷⁵ We represented eye movements towards the left of the animal as negative speed, and rightward movements as positive.

Full field drifting gratings were shown on the 3-screen setup. To effectively elicit OKR in mice, we used a slightly different set of parameters for drifting gratings and plaids. Specifically, for gratings, we used SF = 0.16 cpd and TF = 1.5Hz. For 45° and 20° plaids, we used SF = 0.16 cpd, TF1 = 0.3 Hz, TF2 = 1.2 Hz for component gratings. For 135° and 160° plaids, we used the same SF = 0.16 cpd, TF1 = 0.4 Hz, TF2 = 1.6 Hz for component gratings. These parameters made the VS motion speed for plaid and grating stimulus to be close, with 9.28, 8.92, 8.42, 7.70, 9.38 deg/s for 20°, 45°, 135°, 160° plaids and single gratings, respectively. Each stimulus was presented for 10 seconds with a 3s inter-stimulus-interval and repeated for 15 times in a pseudorandom order.

QUANTIFICATION AND STATISTICAL ANALYSIS

Direction selectivity

We quantified the direction selectivity for each neuron/multi-unit cluster with global direction selectivity index (gDSI) as $gDSI = \frac{\sum R_{\theta} e^{i\theta}}{\sum R_{\theta}}$, where R_{θ} is the response magnitude to stimulus direction θ in radians. The preferred direction was calculated as the angle of the vector summation (i.e., $\theta_{pref} = \arg(\sum R_{\theta} e^{i\theta})$).

OKR gain

Because the stimulus speed was different across gratings and plaids, we calculated the OKR gain to compare between the responses to different stimuli, $OKR\ Gain = \frac{speed_{eye}}{speed_{stim,VS}}$. Here, we used the stimulus speed at the VS direction as the denominator because we observed that the OKR direction was generally consistent with the VS direction. The preferred direction of OKR was calculated as the angle of the vector summation of the OKR Gain tuning curve. OKR gain range was calculated by the maximum OKR gain minus the minimum.

Probabilistically Constrained Vector Sum Model

The individual grating's prediction function was a probability density function (pdf, $w(x)$) approximated as a normalized product of two functions as follows:

$$v(x) = t \cdot \tanh\left(\frac{s \cdot \sec^2(x)}{t}\right)$$

$$u(x) = \frac{1 - \alpha}{360} + \frac{\alpha}{b - a} \frac{1}{1 + \exp\left(-\frac{g(x)}{c}\right)} \frac{1 - \exp\left(-\frac{1}{c}\right)}{1 + \exp\left(\frac{g(x) - 1}{c}\right)}$$

$$w(x) = \frac{v(x) \cdot u(x)}{\int v(x) \cdot u(x) dx}$$

where x is the direction; $v(x)$ applies a soft threshold on the $\sec^2(x)$ function, with s controlling the softness and t being the threshold; $\sec^2(x)$ is the squared secant function; $u(x)$ is a mollified uniform distribution,⁷⁶ where a and b are the lower and upper bounds of the probable direction range given a single grating (i.e. $0^\circ + \delta$ and $180^\circ - \delta$ for a grating moving towards 90° direction, δ being a small offset of two bounds); c is the parameter controlling the slope of the transition between high and low values; α controls the height of the mollified uniform distribution. $g(x)$ is defined as: $g(x) = \frac{x - a}{b - a}$. The normalization is done so that the function properly represents probability. The joint prediction was also a pdf obtained by normalizing the product of two pdfs of individual gratings so that its area-under-curve is 1.

We chose the above functions to estimate the joint prediction for the following reasons. First, given a drifting grating, one half of the full range of directions (covering 180° of the full 360° cycle) is compatible while the other half is not (Figure 2E, yellow shades). Thus, mollified uniform distribution was used to approximate it so that the joint prediction is of high probability for half of the directions but low for the other half. Second, the end point of all the compatible motion vectors of a drifting grating can be represented by its constraint line as in IOC (Figure S2A). In theory, every motion along the constraint line is equally probable (Figures 2E, gray arrows and S2A). If one randomly samples on the constraint line and finds the corresponding motion directions, directions that are away from the grating direction are more likely to be sampled. This relationship between motion directions (x) and their corresponding positions on the constraint line (y) can be described by $y = c \cdot \tan(x)$, where c is the speed of the grating (Figure S2A). The probability of each direction can be approximated by the derivative of y , which is a squared secant function as in $v(x)$. The squared secant function is further applied a soft threshold so that the probability does not go to infinite. This $v(x)$ function contributes to the 'M' shape in the prediction curves (Figures 2E and S2). Note that our main conclusions do not depend on the 'M' shape in the function. If $v(x)$ is substituted with a constant (i.e. making the 'M' flat), the model generates similar predictions that can still explain our data. Here, we use the full form of the function for theoretical accuracy.

The likelihood by VS is modelled with von Mises distribution:

$$L(x) = \frac{\exp(\kappa \cos(x - \mu_{VS}))}{2\pi I_0(\kappa)}$$

where x is the direction; κ is the shape parameter; μ_{VS} is the VS direction; I_0 is the modified Bessel function of order zero.

The final joint pdf, which we propose is what SC neurons represents, is the product of the joint prediction and the likelihood. We kept the joint pdf unnormalized because we found weaker SC responses to plaids with larger cross angles (Figures 2, 3, and 4, example neurons). For these plaids where the VS falls far outside the "likely" range given by the joint prediction, neuronal responses may resemble those elicited by non-directional stimuli.